

Combined Cycle Power Plant Performance Dashboard

Analysis of environmental factors affecting electrical power output using data analytics and machine learning.

454.34

Average Power Output

495.76

Maximum Power Output

420.26

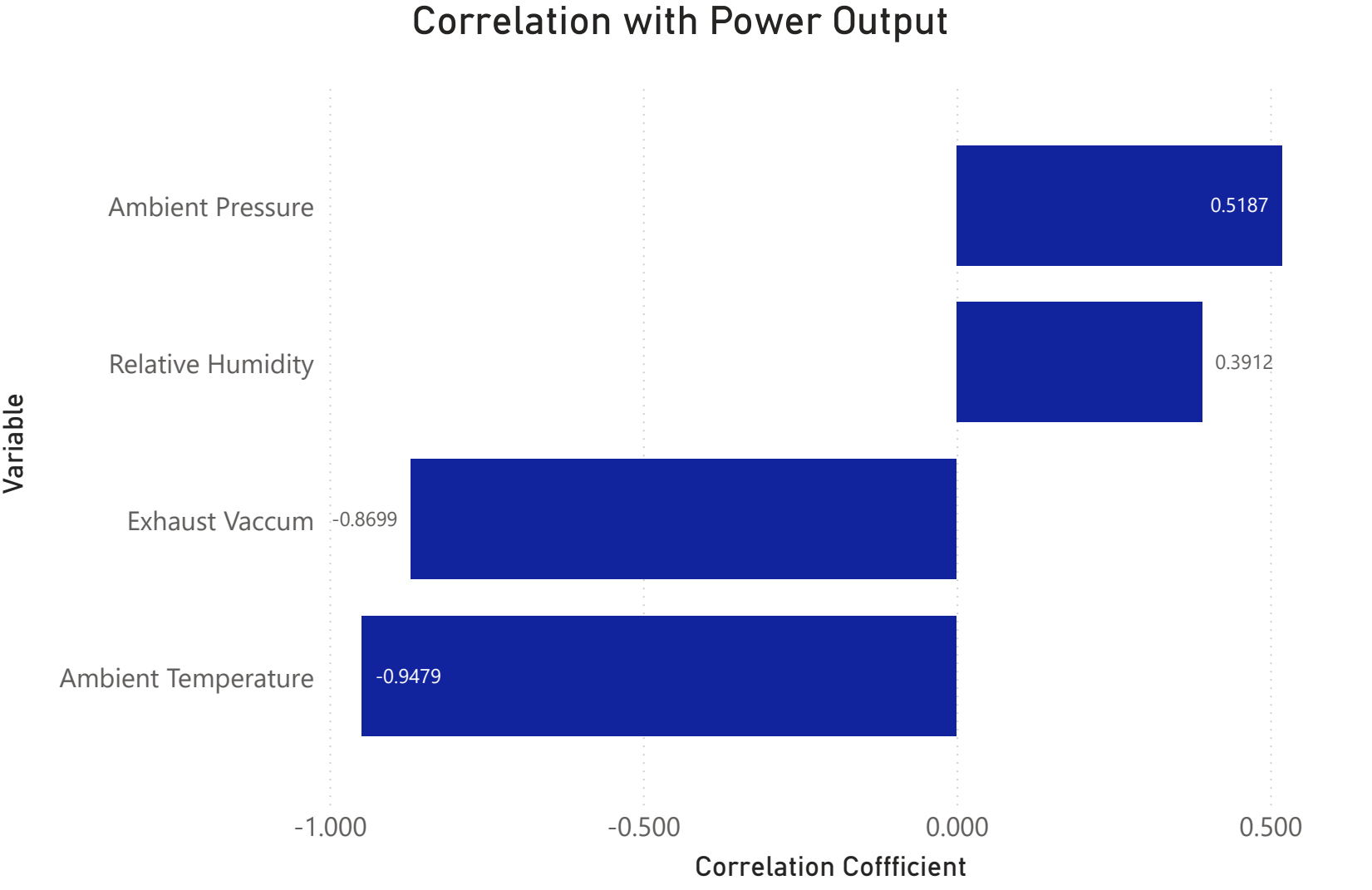
Minimum Power Output

3.62

MAE

0.93

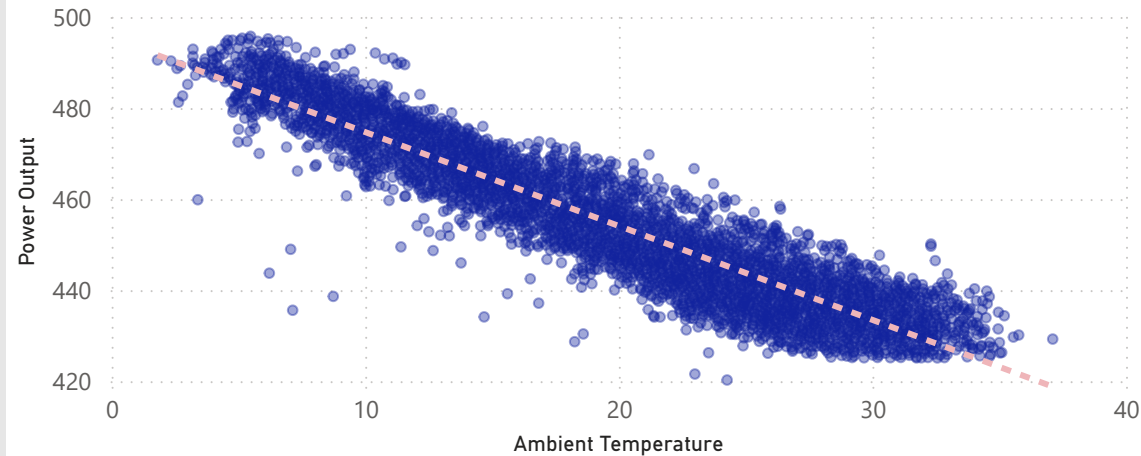
R² Score



How environmental conditions influence electrical power

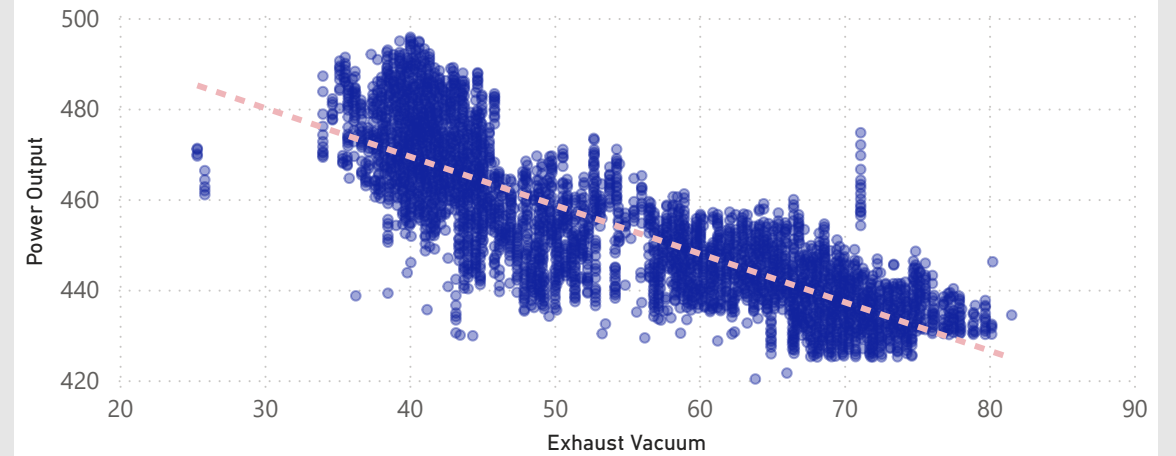
Ambient Temperature vs Power Output

Correlation: -0.9479



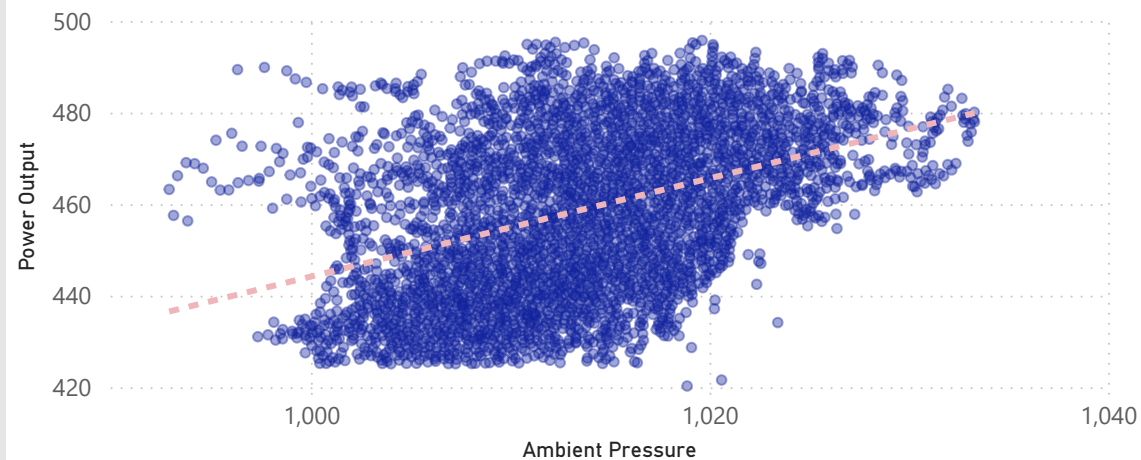
Exhaust Vacuum vs Power Output

Correlation: -0.8699



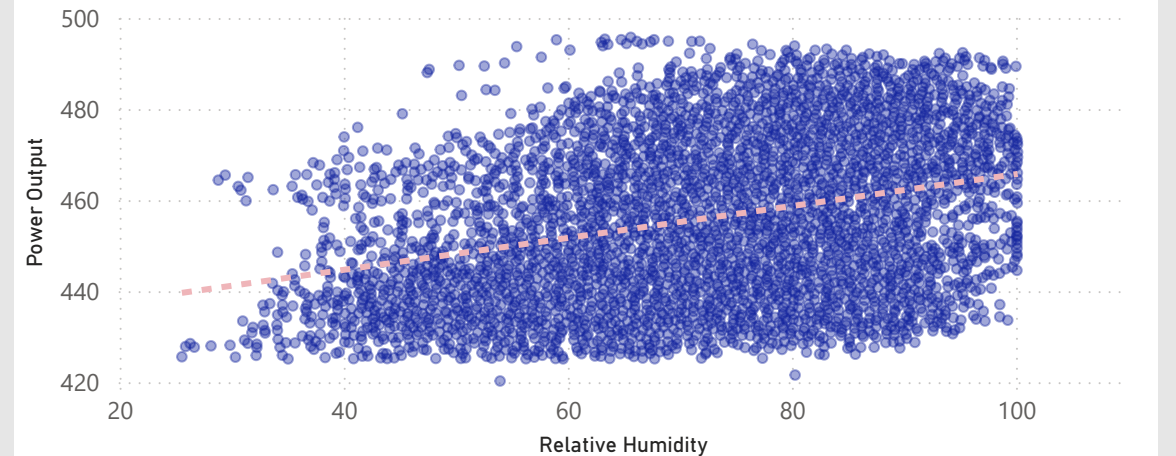
Ambient Pressure vs Power Output

Correlation: +0.5187



Relative Humidity vs Power Output

Correlation: +0.3912





Evaluation of machine learning predictions against power output.

4.59

RMSE

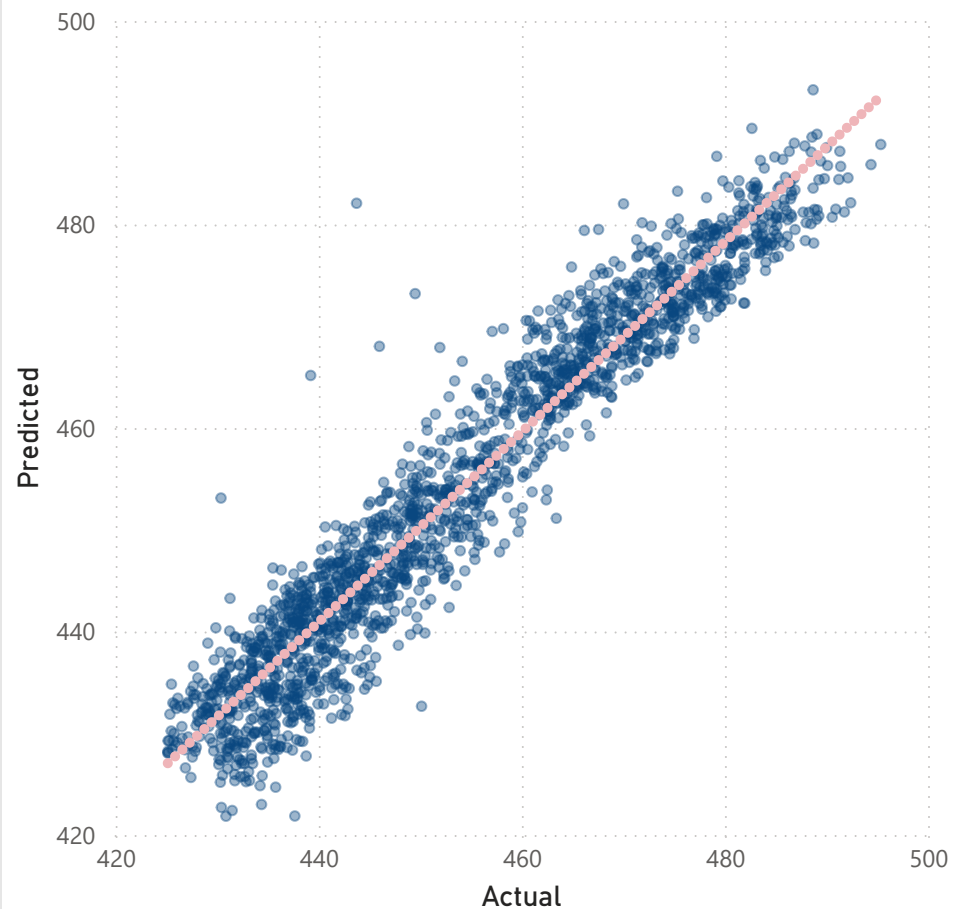
3.62

MAE

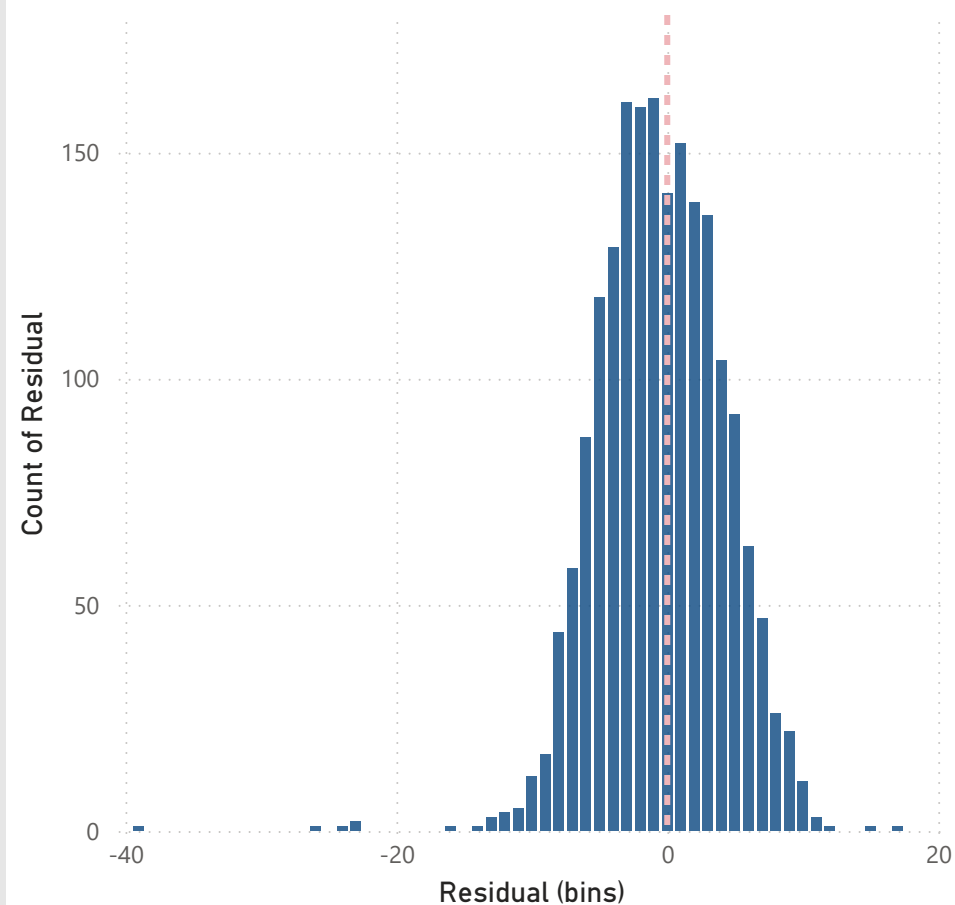
92.84%

R^2 Score

Actual vs Predicted



Residual Distribution



Translate analytical and machine learning findings into actionable business insights.

Business Insights & Recommendations:

Data-driven findings to support power generation performance monitoring and operational decision-making.

Key Finding:

- Ambient Temperature is the strongest factor associated with Power Output, showing a very strong negative relationship ($r = -0.9479$).
- Exhaust Vacuum is the second strongest factor, also showing a strong negative relationship with Power Output ($r = -0.8699$).

Predictive Model Performance:

- The machine learning model achieved an R^2 score of 92.84%, indicating that the model explains a large proportion of the variation in electrical power output. The model produced an MAE of 3.62 MW and an RMSE of 4.59 MW, indicating relatively small prediction errors compared with the overall range of power output.

Business Interpretation:

- The analysis indicates that environmental operating conditions have a significant relationship with electrical power generation. Higher ambient temperatures and exhaust vacuum levels are associated with lower power output, while ambient pressure shows a moderate positive relationship. Relative humidity has a weaker individual relationship with power generation. These findings suggest that temperature and exhaust vacuum should receive greater attention when monitoring plant performance.

Translate analytical and machine learning findings into actionable business insights.

Business Recommendations:

1. Monitor ambient temperature closely:

Since ambient temperature has the strongest relationship with power output, it should be incorporated into routine performance monitoring and operational analysis.

2. Monitor exhaust vacuum as a key operational factor:

The strong negative relationship between exhaust vacuum and power output suggests that changes in this variable should be investigated when unexpected reductions in generation occur.

3. Use multiple environmental variables together:

Power generation is influenced by several environmental conditions. Monitoring temperature, exhaust vacuum, pressure, and humidity together can provide a more complete understanding of operating performance than analyzing individual variables in isolation.

4. Use predictive analytics for performance monitoring:

The model's 92.84% R^2 score suggests that machine learning can provide useful estimates of expected power output. Comparing predicted output with actual output could help identify unusual operating conditions that require further investigation.

5. Integrate the analysis into operational dashboards:

The Power BI dashboard can be used as a monitoring tool to track environmental conditions, power output, prediction accuracy, and residual patterns, supporting faster identification of potential performance deviations.